

Candidate 11141

(2)

(a)

(i) This statement is TRUE: retained earnings are decreased by paying dividends to shareholders. Retained earnings are the sum of all net incomes (revenue less expenses) over the company's life, less any dividends distributed to shareholders. For example, assume TESLA had retained earnings of £200m in 2018, then in 2019 made £800m revenue and incurred £700m in expenses. Thus 2019 profit would equal £100m. Adding this to the retained earnings from 2018 would give a figure of £300m. However, TESLA decided to pay shareholder dividends of £50m. Thus, its retained earnings at the end of 2019 would be £250m.

(ii) This statement is TRUE: unearned, or deferred, revenue is recorded as a liability. In line with accrual accounting principle, revenue is only recognised when it is earned, meaning the product or service has been delivered to and accepted by the customer. Until this point, revenue is unearned and a liability, because it represents an obligation which will incur expenses in the future. For example, if ODEON generate £30,000 in revenue from cinema ticket sales, this revenue is unearned until the movie screening has been shown.

(iii) This statement is TRUE: salaries expense reduces retained earnings, which are the sum of all net incomes (revenue minus expense), less dividends paid. Salaries expense decreases net income, thus decreasing retained earnings.

(iv) This statement is FALSE: prepaid rent is recorded as an asset. All prepaid expenses are recorded as assets until they are used, because they are resources that will generate future income for the company. Shown below is a company for prepaid for 12 months of rent for £50,000 in cash. The prepaid rent is an asset which is debited, and we credit cash.

(Debit) Prepaid rent	£50,000	
(Credit) Cash		£50,000

Then, halfway through the rent's term, the journal entries would look like this:

(Debit)	£6,000	
(Credit) Prepaid rent		£6,000

(b)

Using the indirect method – start from net income and adjust the effect of all non-cash operations to calculate the net cash

+ depreciation of 30k

- Cash increase of 300k and A/R increase of 20k
- Decrease of A/P of 15k
- Gains on sale of land of £110k

Cash flow from investing activities – investments and long-term assets

➔ Increased from selling land for £110k

Financing activities – changes in long-term liabilities and equity

- ➔ Increase from issuing equity £70k
- ➔ Decrease from paying £10k dividends

MST Corp's income for 2018 was £150k (unfinished)

(c)

In line with accrual accounting principles, managers should only recognise revenue when the product generating the revenue has been delivered to and accepted by the customer. At this point, the revenue becomes “earned”. Most companies expect some sales returns, so to adhere to accrual accounting principles they create an estimate of sales returns, perhaps based on historical data, and credit this amount into the deferred/unearned revenue account. An example of this is shown below. Here, the company sells £200,000 of goods which it holds in inventory at a cost of £100,000. Based on historical data, the company estimates that 15% of these sales will be returned after the current accounting period. This means the company’s revenues increase by £170,000.

Recording the sale:

(Debit) Cash	£200,000	
(Credit) Sales Revenue		£170,000
(Credit) Deferred Revenue		£30,000

Recording the cost of the sale:

(Debit) COGS	£100,000	
(Debit) Return Inventory Asset	£85,000	
(Credit) Deferred Revenue		£15,000

However, the manager of this company could commit fraud by purposefully and grossly underestimating the number of sales returns. For example, using an estimate of 5% would result in sales revenue being credited by £190,000, with just £10,000 being recorded as deferred revenue. This would artificially inflate revenues for this accounting period by £20,000. This is shown below.

Fraudulently recording the sale:

(Debit) Cash	£200,000	
(Credit) Sales Revenue		£190,000
(Credit) Deferred Revenue		£10,000

Fraudulently recording the cost of the sale:

(Debit) COGS	£100,000	
(Debit) Return Inventory Asset	£95,000	
(Credit) Deferred Revenue		£5,000

The extra £20,000 of revenue recorded for this accounting period will inflate the company’s net income, thus its earnings per share. The motivation to do so is that earnings per share is used by investors to determine the financial health of a company; if earnings per share rises, the company’s stock becomes more attractive and will likely rise in price. This could be a strong motivator for this manager to commit this fraud, especially if her pay is linked to the company’s stock price. It is worth noting that managers can also commit fraud by overestimating sales returns to therefore under report

revenue. This could be done for tax avoidance purposes, though Boisseau-Sierra (2020) notes this is more easily achieved by using offshore tax havens, for example.

A second example of a journal entry which could be made to commit fraud relates to revenue recognition for bundled products. In my previous example, I explained that the portion of sales expected to be rejected by customers generates unearned, or deferred revenue. Similarly, where a company sells a product bundled with a service, after the product is accepted it is still possible that the accompanying service is an unearned revenue.

If Sony sells its PlayStation4, this might be bundled with a service such as providing online gaming platforms for customers. This is shown below, where the bundle costs £500 and if sold separately, the online gameplay service costs £110.

		If Sold Separately:	Allocation	Bundled Price
Gaming Console	Delivered	£390	£390	£500
Online gameplay service	Undelivered	£110	£110	

(Debit) Cash	£500	
(Credit) Console sales revenue		£390
(Credit) Deferred revenue (online gaming service liability)		£110

Here, the portion of sales revenue generated from the console credits sales revenue, whereas the obligation to provide online gaming services is a liability and therefore currently unearned revenue. Similarly to my previous example, Sony's managers could commit fraud by artificially inflating the proportion of the bundle's value assigned to the console, in order to increase sales revenue and decreased deferred revenue. To illustrate, increasing the value allocated to the gaming console from £390 to £450 would increase revenues for this accounting period by £60 per unit sold. This would thus boost net income and earnings per share for this period. This is fraudulent because if sold separately the gaming console would bring in £390 in revenue.

Fraudulently recording the sales:

		If Sold Separately:	Allocation	Bundled Price
Gaming Console	Delivered	£390	£450	£500
Online gameplay service	Undelivered	£110	£50	

(Debit) Cash	£500	
(Credit) Console sales revenue		£450
(Credit) Deferred revenue (online gaming service liability)		£50

To avoid revenues being recorded early in this manner, companies apply the IFRS 15 standard. This is a five step model which begins with identifying the contract with a customer, identifying the separate performance obligations in the contract (in this case, providing the games console and providing the online gaming service), determining the transaction price, allocating the transaction

price to each of the performance obligations, and then recognising each revenue when the company has satisfied the corresponding performance obligation.

4.

(a)

Normally, the price of a bond is given by:

$$P_0 = \frac{C}{YTM} \times \left(1 - \frac{1}{(1 + YTM)^t}\right) + \frac{FV}{(1 + YTM)^t}$$

Where C is the coupon, YTM is the bond's yield to maturity, t is the number of time periods in which a coupon is paid, and FV is the face value or par value of the bond, the capital sum repaid to the bondholder at the bond's maturity date.

Here, we have a zero-coupon bond, meaning C = 0. Hence:

$$P_0 = \frac{FV}{(1 + YTM)^t}$$

$P_0 = £82$ and $FV = £100$. We also know that $t = 5$. Hence:

$$£82 = \frac{£100}{(1 + YTM)^5}$$

Solving for YTM:

$$£82 (1 + YTM)^5 = £100$$

$$(1 + YTM)^5 = \frac{£100}{£82}$$

$$(1 + YTM) = \sqrt[5]{\frac{£100}{£82}}$$

$$1 + YTM = 1.040488368$$

$$YTM = 0.040488368$$

$$YTM = 4.0488368\%$$

$$YTM \approx 4.05\%$$

Checking the answer:

$$\frac{£100}{(1.0405)^5} = £81.99$$

This means that a bondholder who purchases the bond for £82 and holds it until its maturity date will receive a rate of return of 4.05%.

(b)

$$P_0 = \frac{FV}{(1 + YTM)^t}$$

In this situation, the bank believes a “fair” yield-to-maturity on its bond is 3.55%. This could be for multiple reasons, one being that AI Logistics has a strong credit rating and is therefore a lower-risk investment for the bondholder. In this situation, the bank may feel it is entitled to borrow at lower rates, hence it wishes to price the bond in a way that reduces the YTM to 3.55%. Assuming the par value of the bond (FV) remains at £100, and $t = 5$, we can solve this equation for P_0 :

$$P_0 = \frac{£100}{(1.0355)^5}$$

$$P_0 = £83.99423594$$

$$P_0 \approx £84$$

Thus, issuing the bond for £84 will mean that an investor who holds the bond until its maturity will receive a rate of return of 3.55%. This result is intuitive; if the par value of the bond has remained constant, but the price paid has increased, naturally we would expect the rate of return to fall.

3. Completed based on Dr. Cole's instruction (Week 8 Q&A lecture) that where both answers are completed the highest scoring answer will count towards the overall mark.

$$D_5 = £4.50, g = 0.025, r = 0.09$$

(a)

The price of the share today is equal to the present value of the share from years 1 to 4, plus the present value of share from year 5 onwards. This is shown in the equation below:

$$P_0 = PV_{1-3} + PV_{4-\infty}$$

Here, the dividend is zero in years 1-4, hence the stock will have a value of zero. From year 5 onwards, the stock is a perpetuity with constant growth. So, we need to calculate the value of this perpetuity using the following formula:

$$P_{4-\infty} = \frac{D_5}{r - g}$$

$$P_4 = \frac{4.50}{0.09 - 0.025}$$

$$P_4 = £69.23076923$$

However this is P_4 in year 4. We need to discount this figure to find NPV_4 :

$$PV_4 = \frac{P_4}{(1+r)^4}$$

$$PV_4 = \frac{£69.23076923}{(1 + 0.09)^4}$$

$$PV_4 = £49.044$$

(b)

$$P_0 = PV_1 + PV_{2-\infty}$$

Again, as no shares are paid in year 1 or 2

$$P_2 = \frac{D_3}{r - g}$$

$$P_2 = \frac{4.50}{0.09 - 0.025}$$

$$P_2 = \frac{4.50}{0.09 - 0.025}$$

$$P_2 = £69.23076923$$

$$PV_2 = \frac{£69.23076923}{(1 + 0.09)^2}$$

$$PV_2 = £58.27015338$$

$$PV_2 = £58.27$$